

Scalability & Future Plan

Date	30 September 2026
Team ID	SWTID-2026-6111
Project Name	FIT BUDDY AI
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1 1	Local deployment	Designed and tested on a local FastAPI/Uvicorn server; large-scale access is not documented.	High
2 2	SQLite data storage	User and plan data are stored in SQLite; larger data loads may require a more scalable storage approach.	Medium
3 3	Production-scale security controls not specified	The project documents environment-based API-key storage, but does not specify full production security controls.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1 1	User Load User Load	Local FastAPI/Uvicorn deployment	Move to cloud deployment and scale application instances as user demand grows.
2 2	Data Storage Data Storage	SQLite with SQLAlchemy ORM	Evaluate migration to a scalable database when data volume and concurrent access increase.
3 3	Performance Performance	Gemini Pro for plans; Gemini Flash for fast tips	Use scalable deployment and optimize AI/API request handling as usage grows.

4 4	Security Security	Gemini API key stored through environment variables/.env	Add production-grade secret management and access controls before wider deployment.
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Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2 Phase 2	Cloud deployment of the application	Future / To be decided	Improves availability and supports growth beyond the current local deployment.
Phase 3 Phase 3	Scalable data storage and expanded user handling	Future / To be decided	Supports larger data volumes and more concurrent users.
Phase 4 Phase 4	Extended AI-powered fitness support and optimization	Future / To be decided	Evolves FitBuddy with additional capabilities while maintaining personalized plans.